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Studierichting: Informatica
Oefeningenreeks 4A nr. 47.2

$$\int \frac{3x-1}{(x^2+4x+5)^3}dx = \int \frac{3x-1}{((x+2)^2+1)^3}dx$$

Stel $t = x+2 \Rightarrow dt = dx \Rightarrow x = t-2$:

$$\int \frac{3(t-2)-1}{(t^2+1)^3}dt = \int \frac{3t-7}{(t^2+1)^3}dt$$
$$= 3\int \frac{t}{(t^2+1)^3}dt - 7\int \frac{1}{(t^2+1)^3}dt$$

Eerste integraal:

$u = t^2+1 \Rightarrow du = 2tdt$:

$$3\int \frac{t}{(t^2+1)^3}dt = \frac{3}{2}\int u^{-3}du = \frac{3}{2}\cdot\frac{u^{-2}}{-2} = -\frac{3}{4(t^2+1)^2}$$

Tweede integraal formule:

$$I_n = \int \frac{dt}{(t^2+1)^n} = \frac{t}{2(n-1)(t^2+1)^{n-1}} + \frac{2n-3}{2(n-1)}I_{n-1}$$

Voor $n=2$:

$$I_2 = \frac{t}{2(t^2+1)} + \frac{1}{2}\operatorname{Bgtan} t$$

Voor $n=3$:

$$I_3 = \frac{t}{4(t^2+1)^2} + \frac{3}{4}I_2 = \frac{t}{4(t^2+1)^2} + \frac{3t}{8(t^2+1)} + \frac{3}{8}\operatorname{Bgtan} t$$

Samen:

$$\int \frac{3x-1}{(x^2+4x+5)^3}dx$$
$$= -\frac{3}{4(t^2+1)^2} - 7\left(\frac{t}{4(t^2+1)^2} + \frac{3t}{8(t^2+1)} + \frac{3}{8}\operatorname{Bgtan} t\right) + c$$
$$= \frac{-3-7t}{4(t^2+1)^2} - \frac{21t}{8(t^2+1)} - \frac{21}{8}\operatorname{Bgtan} t + c$$

Substitutie terug naar x:

$$\int \frac{3x-1}{(x^2+4x+5)^3}dx$$
$$= \frac{-7x-17}{4(x^2+4x+5)^2} - \frac{21(x+2)}{8(x^2+4x+5)} - \frac{21}{8}\operatorname{Bgtan}(x+2) + c$$

References